Linear algebra

LI/LD:

$S_1 V_1 + S_2 V_2 + \cdots + S_n V_n = 0$ } If $\forall i,\ s_i = 0$ is the only soln $\Rightarrow$ LI.
} If some $s_i \neq 0$ gives a soln $\Rightarrow$ v_i's are LD.

one of the vectors can be written as a linear combina^n of the others.

Row operas
- $R_i \leftrightarrow R_j$ — Swap
- $R_i \leftarrow k \cdot R_i$ — scalar multiplica^n
- $R_i \leftarrow R_i \pm k \cdot R_j$ — linear combina^n (makes row LD on other row, i.e. matrix non-invertible)

Row opera^{n}s are invertible.

Matrices
- $A^2 = A$
- $A^2 = 0$

Systems of lin eq^{n}s:

$$\begin{bmatrix} a & b & b_1 \\ c & d & b_2 \end{bmatrix} \longrightarrow \begin{bmatrix} a & b & b_1 \\ 0 & d' & b_2' \end{bmatrix} \xrightarrow{\text{echelon form}} \begin{bmatrix} 1 & b' & b_1' \\ 0 & 1 & b_2' \end{bmatrix} \xrightarrow{\text{row-reduced}}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \xrightarrow[\text{opera's}]{n-1} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22}' & \cdots & \vdots \\ \vdots & \vdots & & \\ 0 & a_{n2}' & \cdots & a_{nn}' \end{bmatrix} \xrightarrow{n-2} \begin{bmatrix} a_{11} & a_{22}' & & a \\ 0 & 0 & a_{33}' & \vdots \\ \vdots & \vdots & \vdots & a_n' \\ 0 & 0 & a_n' & a \end{bmatrix} \longrightarrow \cdots \longrightarrow$$

$(n-1)$ for zeros under 1st pivot
$(n-2)$ (2nd)
$\vdots$
1

$$(n-1) + (n-2) + \cdots + 1 = \frac{(n-1)(n)}{2} = \frac{n^2-n}{2}$$

For row reduced: $(n-1)$ to make these 0 ... $(n-2)$ to make these zero ... $\left(\frac{n^2-n}{2}\right)$ more in total

+ n opera^{n}s to make all pivots 1 (div each row by pivot)

$$\frac{n^2-n}{2} + \frac{n^2-n}{2} + n = n^2 - n + n = n^2$$

For echelon form $\rightarrow \frac{n^2-n}{2}$ opera^{n}s.
For RREF $\rightarrow n^2$ opera^{n}s.
} $O(n^2)$

$Ax = 0$
- unique/trivial → If A has LI col^{m}s
- infinite → If A has LD col^{m}s

$Ax = b$
- No soln → b is NOT a lin comb^n of col^n space of A → Rank(A) < Rank([A|b])
- If there is a soln → b lies in span of A → rank(A) = rank([A|b]) = r
  - unique soln → $r = n$ → only if A is a sq matrix.
  - ∞ soln → $r < n$

Vector spaces:

If u and v belong to vector space V, then $\alpha u + \beta v \in V$.

If V and W are spaces $\rightarrow \dim(V+W) = \dim(V) + \dim(W) - \dim(V \cap W)$

$Ax = b$ → x_n, x_p

Rank(A) + Nullity(A) = n

If we do not have any x_p for $Ax = b$ $\Rightarrow$ Nullity(A) = n, rank = 0

After converting to RREF $\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}\begin{bmatrix} -F \\ I \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\begin{pmatrix} 1 & 0 & 2 & -6 \\ 0 & 1 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ (pivot, free)

soln → $\begin{bmatrix} -F \\ I \end{bmatrix}$ $\begin{bmatrix} -2 & 6 \\ -1 & -7 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ soln → i.e. Null space of A is col space of this

$EA = U$

$A = E^{-1}U = LU$

$Ax = 0$

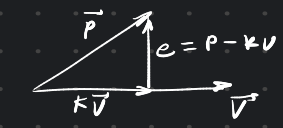
$EAx = 0$ ← soln stays same even after row operations.

$PEAx = 0$ ← even after applying permutation matrix (swapping rows).

Applying E → col space changes. $\text{col}(EA) \subseteq \text{col}(E) \neq \text{col}(A)$

Applying P → col space may / may not change.

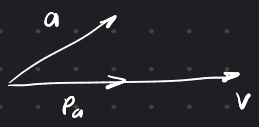
Projections:



Projection of p on v: $\vec{p_v} = v\left(\frac{p^T v}{\|v\|_2^2}\right)$

Projection matrix: $P = \frac{vv^T}{\|v\|_2^2}$

$P\vec{p} = k\vec{v} = \left(\frac{vv^T}{\|v\|_2^2}\right)\vec{p}$



$p_a = P\vec{a} = k\vec{v}$

$Pp_a = PP\vec{a}$ (applying projection matrix twice)

$= P\vec{a} = k\vec{v}$ (makes no difference)

$P^2 = P$

$P^T = \left(\frac{vv^T}{\|v\|_2^2}\right) = \left(\frac{vv^T}{\|v\|_2^2}\right)$

$P^T = P$

P → rank 1 matrix

$v^T v$

$\begin{bmatrix} \quad \end{bmatrix}_{n\times 1}\begin{bmatrix} \quad \end{bmatrix}_{1\times n} = \begin{bmatrix} \quad \end{bmatrix}_{n\times n}$ each col is scalar multiple of $\vec{v}$.